

Lecture 3

Polymerization



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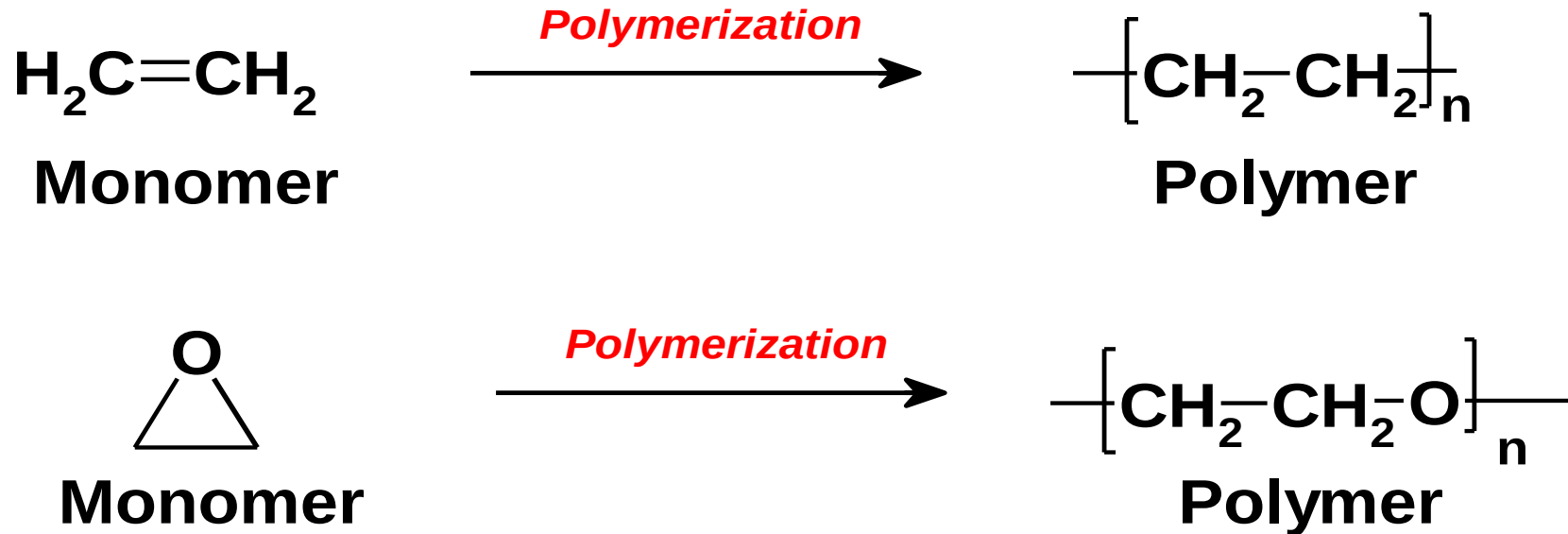
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Polymerization

Polymers

Polymers are large molecules made up of repeating units called **Monomers**. The synthetic process is **Polymerization** in which monomers are joined together to form large, chain like molecules.

Polymers: Macromolecules formed by the covalent attachment of a set of small molecules termed monomers.



Polymerization

The chemical reaction in which high molecular mass molecules are formed from monomers.

There are two basic types of polymerization

- 1) Chain-reaction (or addition) polymerization
- 2) Step-reaction (or condensation) polymerization

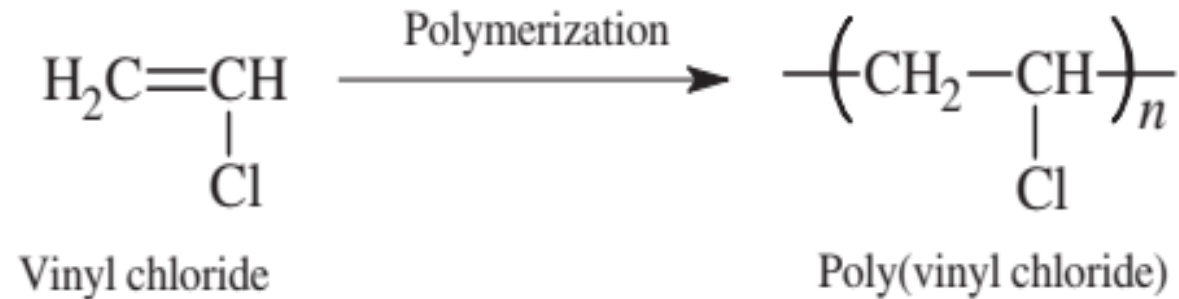
Methods for making polymers

Addition polymerization: monomers react to form a polymer without net loss of atoms.

This type of polymerization is a **three-step process** involving **two chemical components**

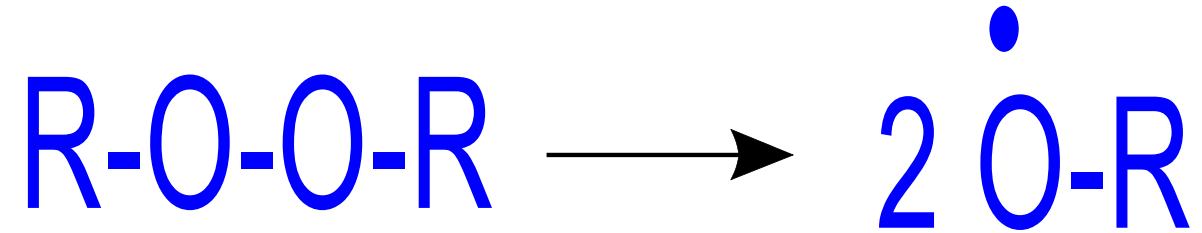
1) Monomer

Can be regarded as one link in a polymer chain. It initially exists as simple units. **In nearly all cases**, the monomers have at least one carbon-carbon double bond. **Ethylene** is one example of a monomer used to make a common polymer.



2) Catalyst

In chain-reaction polymerization, the catalyst can be a **free-radical peroxide** added in relatively low concentrations

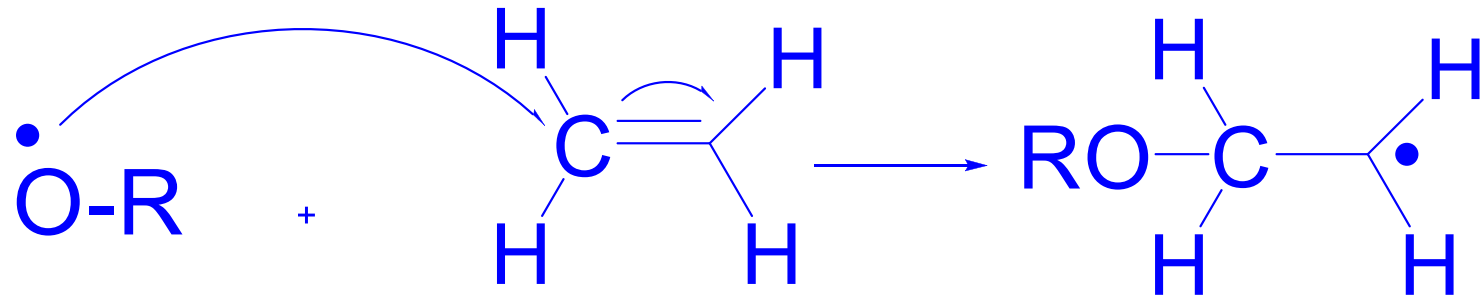


A free-radical is a chemical component that contains a free electron that forms a covalent bond with an electron on another molecule.

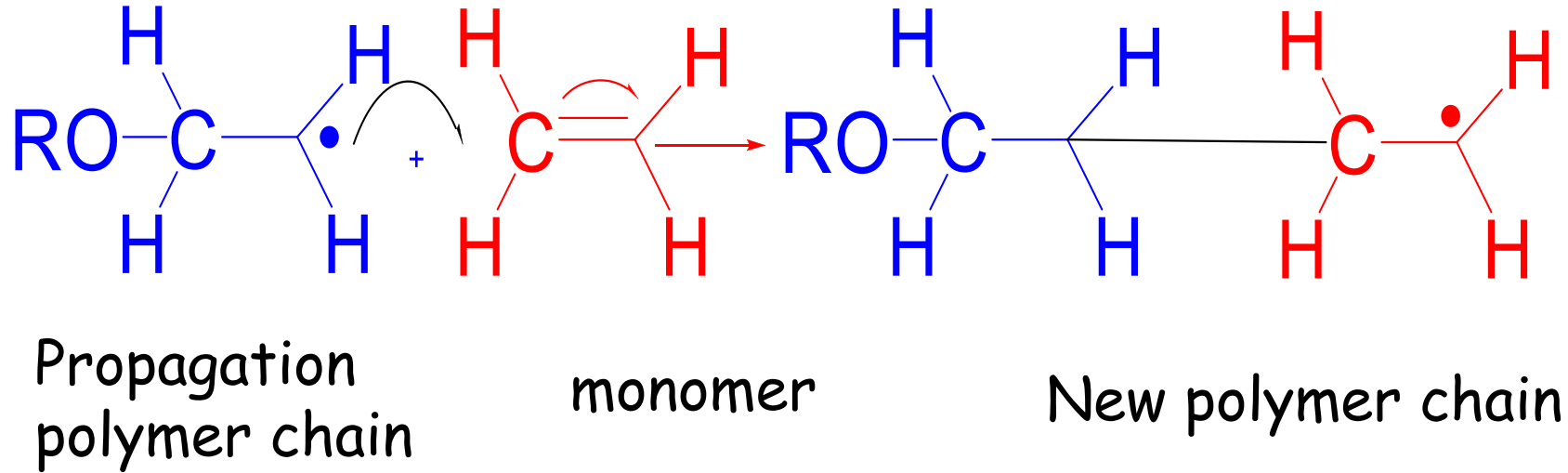
Polymerization Process (Mechanism)

Step 1: Initiation

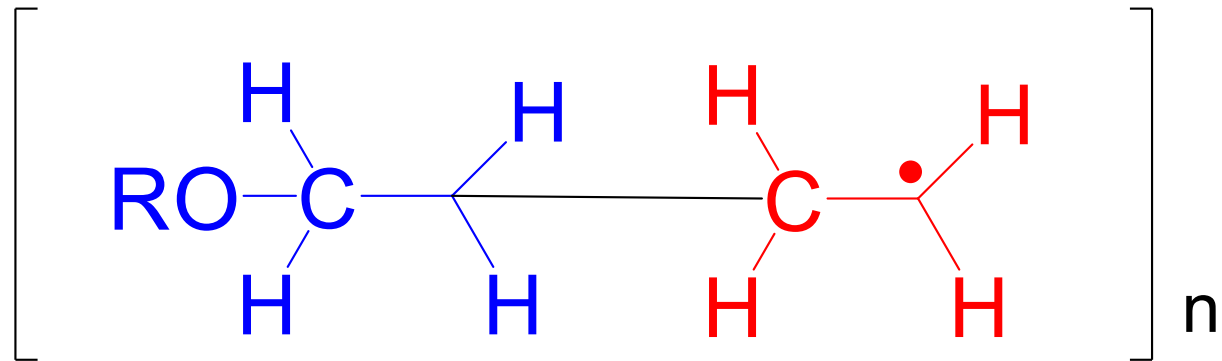
The first step in the chain-reaction polymerization process, initiation, occurs when the free-radical catalyst reacts with a double bonded carbon monomer, beginning the polymer chain. The double carbon bond breaks apart, the monomer bonds to the free radical, and the free electron is transferred to the outside carbon atom in this reaction.



2) Propagation Step



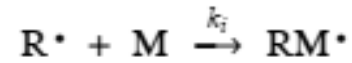
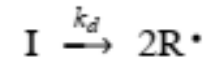
The next step in the process, propagation, is a repetitive operation in which the physical chain of the polymer is formed. The double bond of successive monomers is opened up when the monomer is reacted to the reactive polymer chain. The free electron is successively passed down the line of the chain to the outside carbon atom



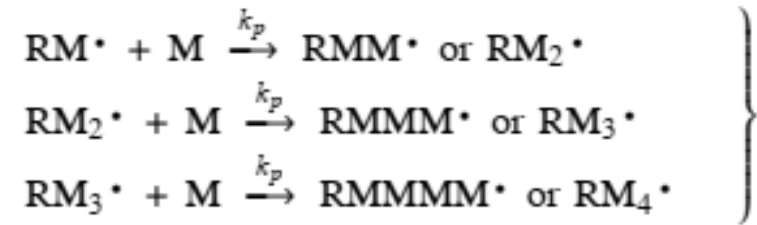
Termination

Termination occurs when another free radical (R-O·), left over from the original splitting of the organic peroxide, meets the end of the growing chain. This free-radical terminates the chain by linking with the last CH₂· component of the polymer chain. This reaction produces a complete polymer chain. Termination can also occur when two unfinished chains bond together.

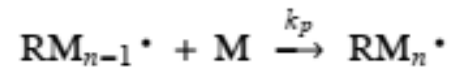
Initiation



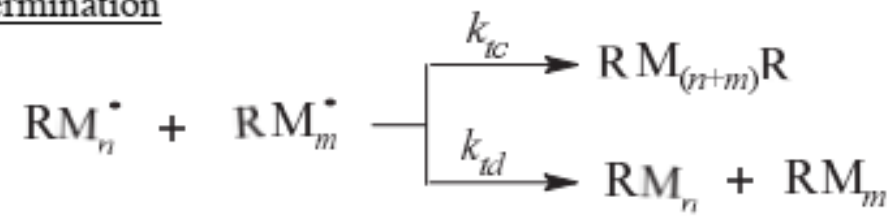
Propagation



In general terms,



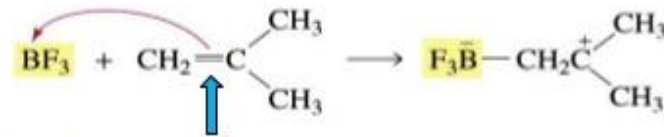
Termination



Addition Polymerization

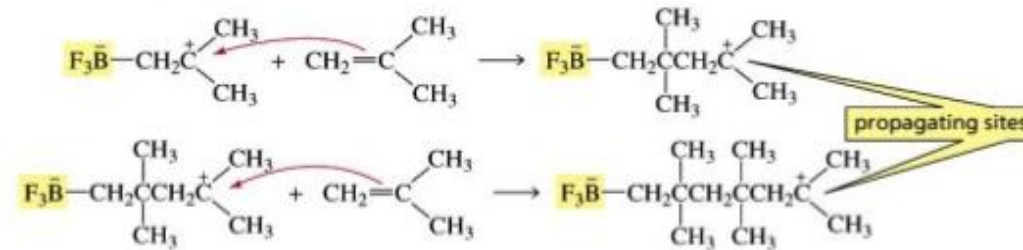
Cationic polymerization: A polymerization where a cation is formed causing a chain reaction. It results in forming a long chain of repeating monomers

chain-initiating step



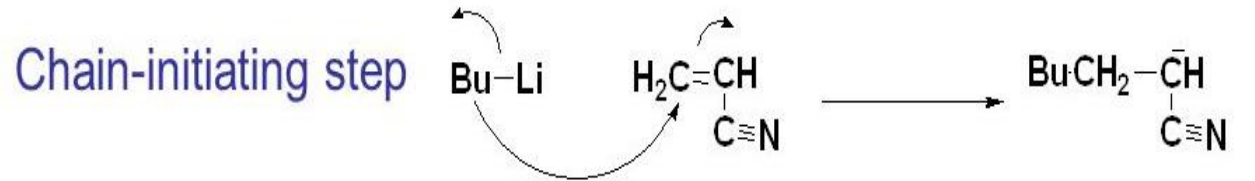
the alkene monomer
reacts with an electrophile

chain-propagating steps

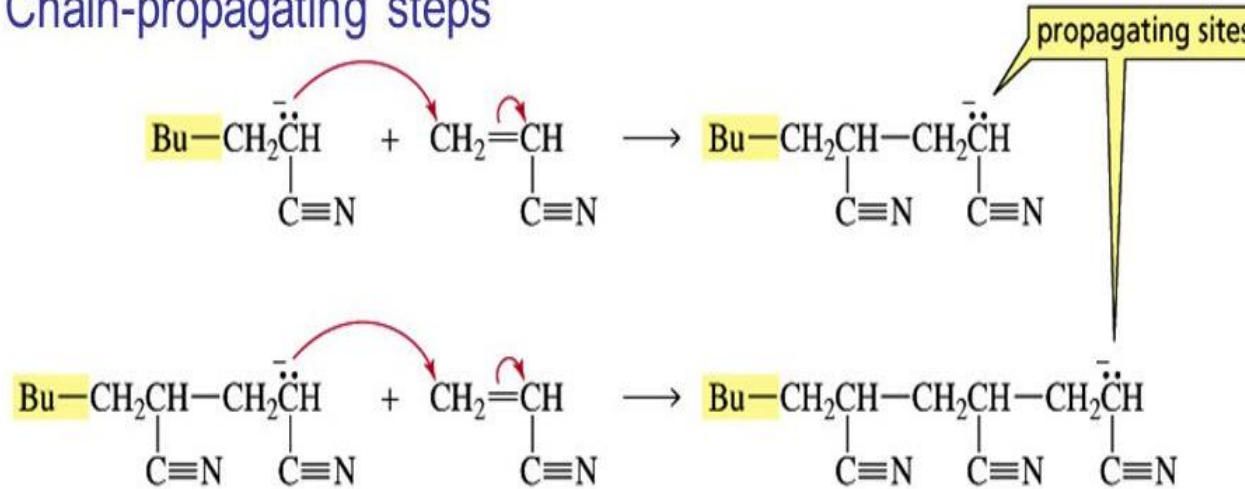


Addition Polymerization

Anionic Polymerization: Involves the polymerization of particularly vinyl polymers with a strong electronegative group to form a chain reaction.

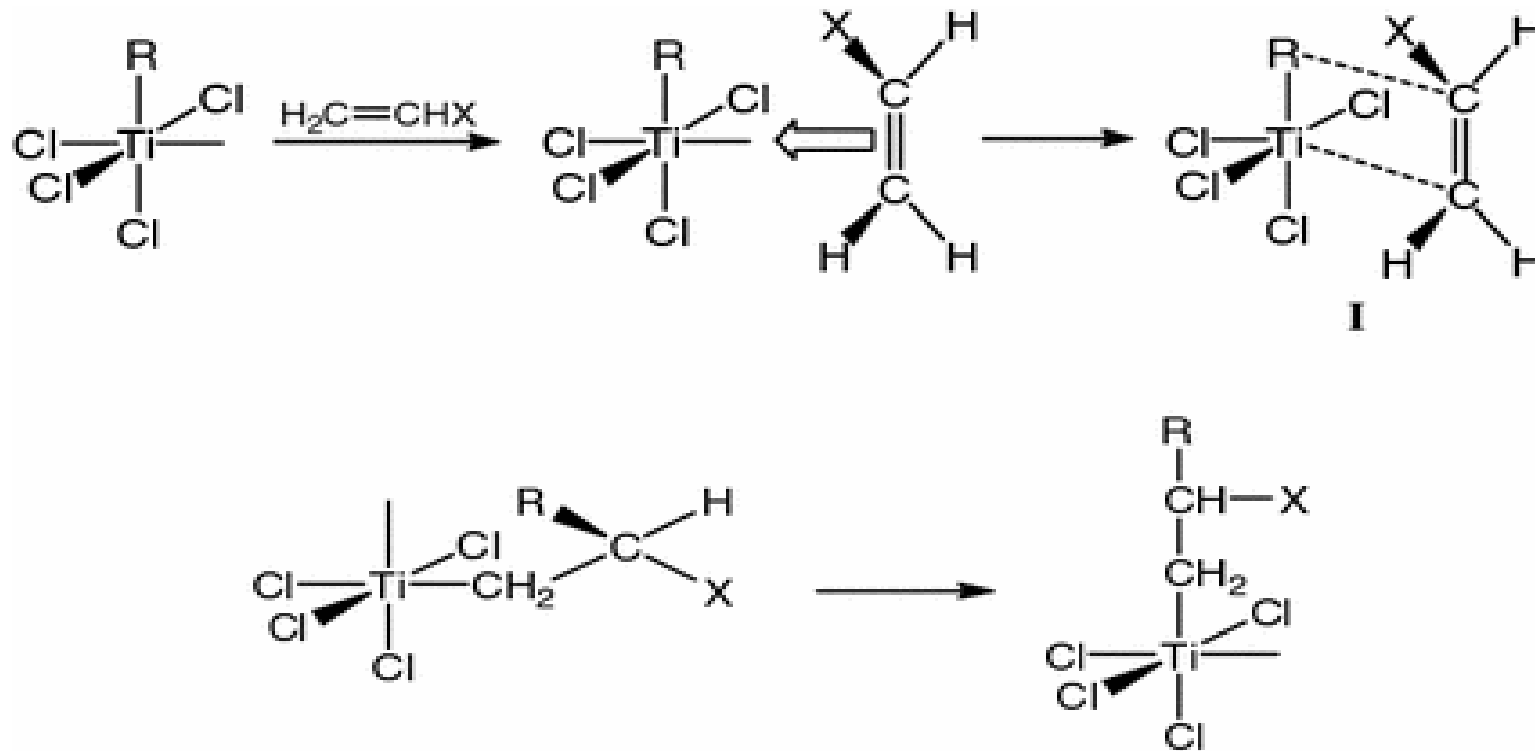


Chain-propagating steps



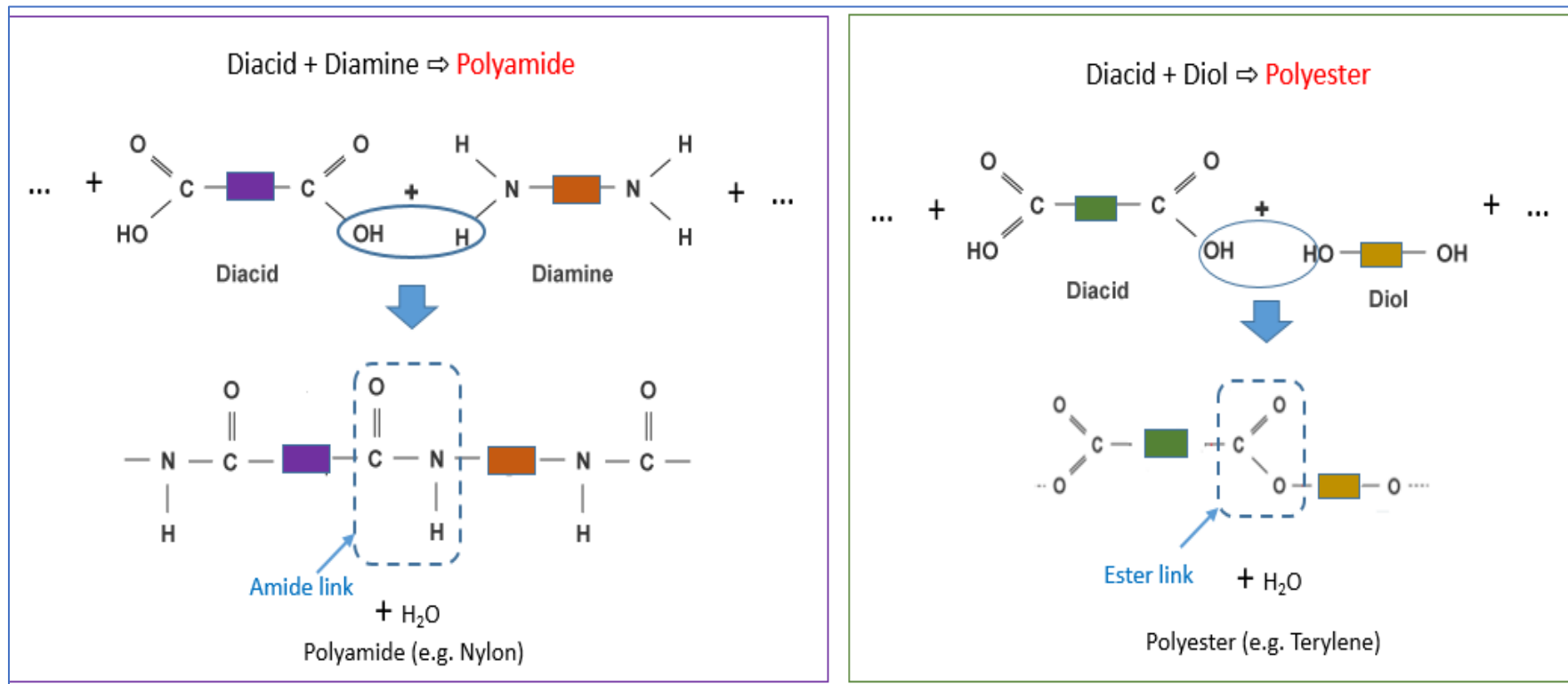
Addition Polymerization

• *Coordination Polymerization*: This method was invented by two scientists Ziegler and Natta who won a Nobel Prize for their work. They developed a catalyst which let us control the free radical polymerization. It produces a polymer which has more density and strength.



Condensation polymerization

Condensation polymers form from the step growth polymerization. Here when molecules of monomers react to form a bond they replace certain molecules. These molecules are the by-product of the reaction. In most cases, this by-product is a water molecule. Polyester and nylon are two common condensation polymers.



Range of Polymers

- Traditionally, the industry has produced two main types of synthetic polymer – plastics and rubbers.
- Plastics are (generally) rigid materials at service temperatures
- Rubbers are flexible, low modulus materials which exhibit long-range elasticity.

Polymer Classifications

Thermoplastics can be softened or melted by heat and reformed (molded) into another shape. Most addition polymers are thermoplastics. The polymer chains are held together by weak interactions (noncovalent bonds) such as :??

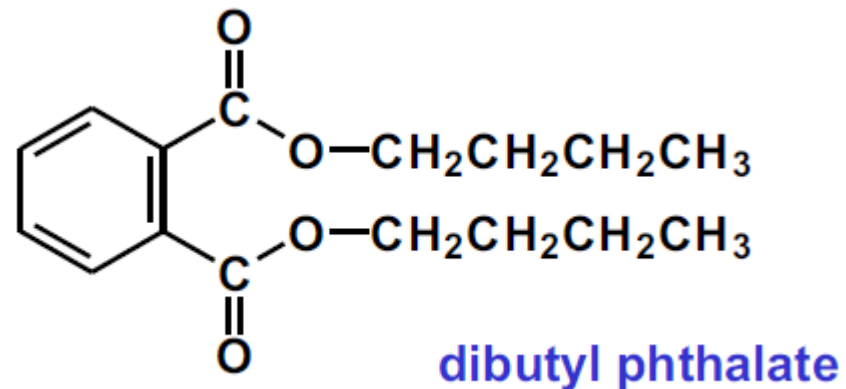
These interactions are disrupted by heating, allowing the chains to become independent of each other. Heating and reforming can be repeated indefinitely (if degradation doesn't occur). This allows recycling.

Most polymers of high molecular weight are quite rigid.

PLASTICIZERS

Can be softened and made flexible by adding plasticizers

The plasticizer separates the individual polymer chains from one another. It acts as a lubricant which reduces the attractions between the polymer chains.



Thermoplastic Polymer

- A **thermoplastic**, or **thermo-softening plastic**, is a plastic material, polymer, that becomes pliable or moldable above a specific temp and solidifies upon cooling.
- thermoplastics may be reshaped by heating Temperature and solidifies upon cooling.

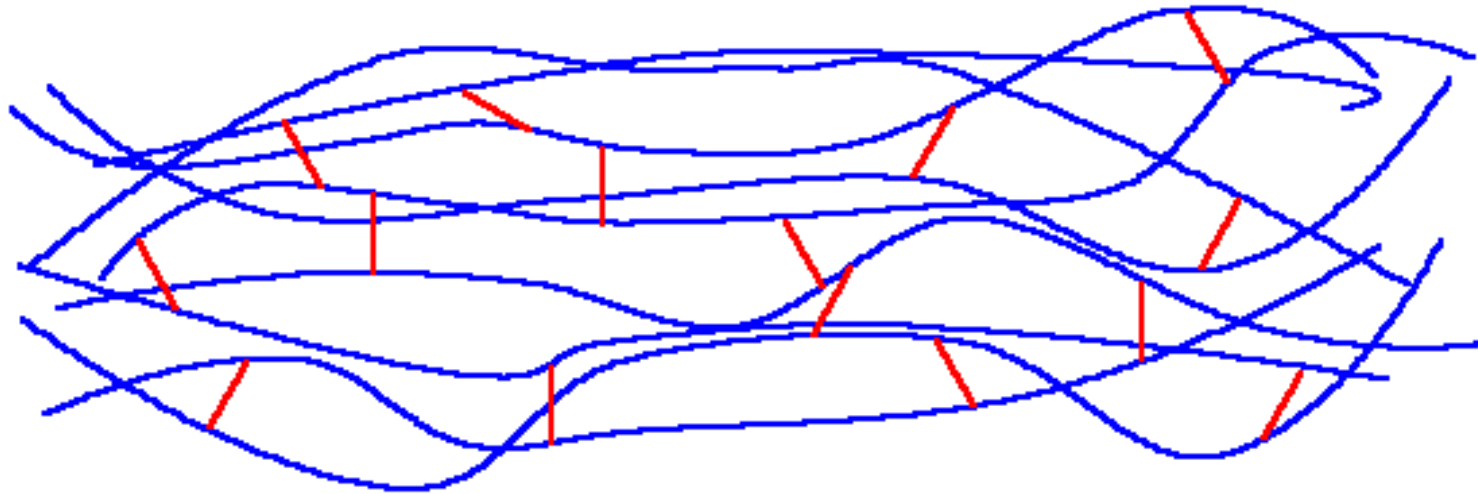


Thermosetting polymers

- Thermosetting polymers have their chains cross linked by covalent bonds.
- The starting materials are placed into a mould to form the desired shape.
- The polymer is then heated (or initiated with uv light) and chemical reactions occur to form the cross links between the chains.
- The resulting three dimensional solid structure cannot then be changed.
- Further heating will not cause the polymer to soften, melt or change shape (unlike thermosoftening polymers).

Thermosetting polymers

- The picture below shows a typical structure for a thermosetting polymer. The red lines represent the cross links between the chains.

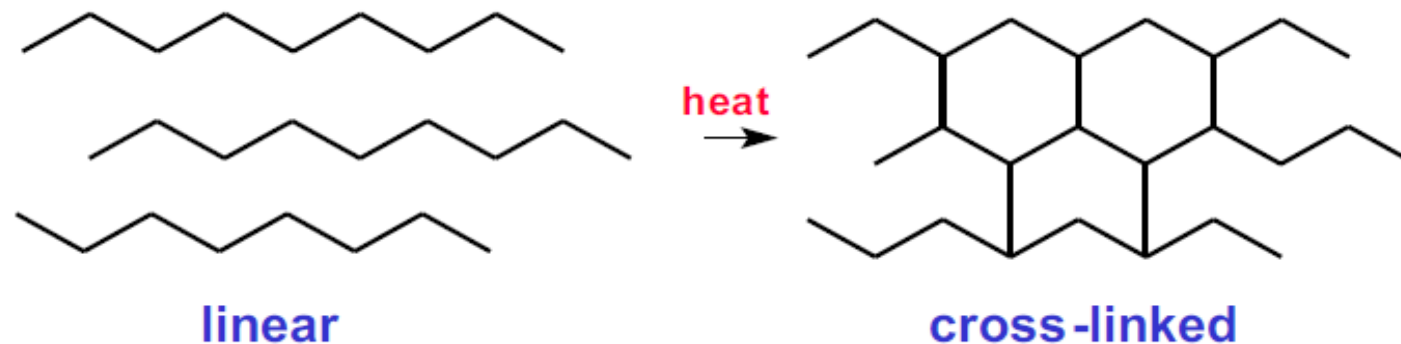


Examples

- Epoxy resins - used as coating materials, caulks, manufacture of insulating materials, etc ...
- Phenolic resins - tool handles, billiard balls, sprockets, insulation, etc ...
- Unsaturated polyester resins - manufacture of plastics reinforced fiberglass commonly known as polyester, fillers, etc ...

Thermoset Plastics

- Thermoset plastics melt initially, but on further heating they become permanently hardened
- Once formed, thermoset plastics cannot be remoulded, and they can not be recycled
- On heating, thermoset plastics become cross-linked (covalent bonds forms between the chains). The cross-linked chains form a rigid network



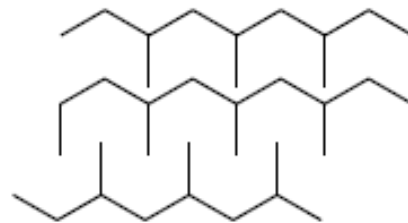
High-density polymers

Linear polymers with chains that can pack closely together. These polymers are often quite rigid.



Low-density polymers

Branched –chain polymers that can not pack together as closely. There is often a degree of cross linking. These polymers are often more flexible than high-density polymers

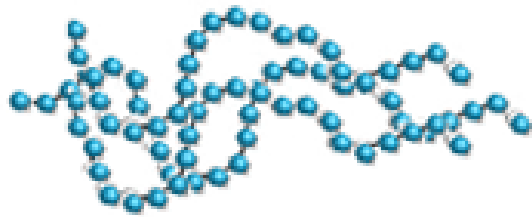


Elastomers

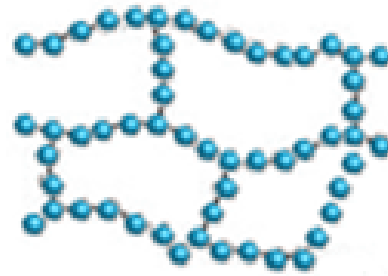
- **Elastomer** materials are those materials that are made of polymers that are joined by chemical bonds, acquiring a final slightly crosslinked structure.



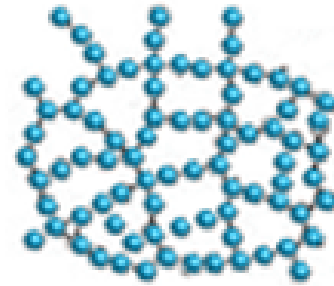
- Natural rubber - material used in the manufacture of gaskets, shoe heels...
- Polyurethanes - Polyurethanes are used in the textile industry for the manufacture of elastic clothing such as lycra, also used as foam, wheels, etc ...
- Polybutadiene - elastomer material used on the wheels or tires of vehicles, given the extraordinary wear resistance.
- Neoprene - Material used primarily in the manufacture of wetsuits is also used as wire insulation, industrial belts, etc ...
- Silicone - Material used in a wide range of materials and areas due to their excellent thermal and chemical resistance, silicones are used in the manufacture of pacifiers, medical prostheses, lubricants, mold, etc ...



Thermoplastic

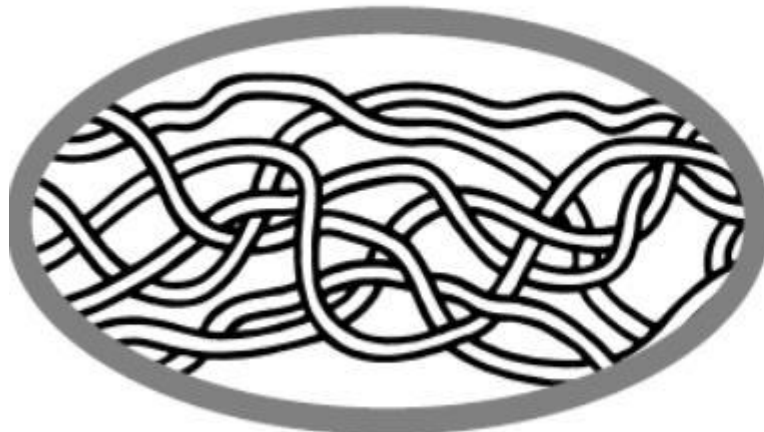


Elastomer



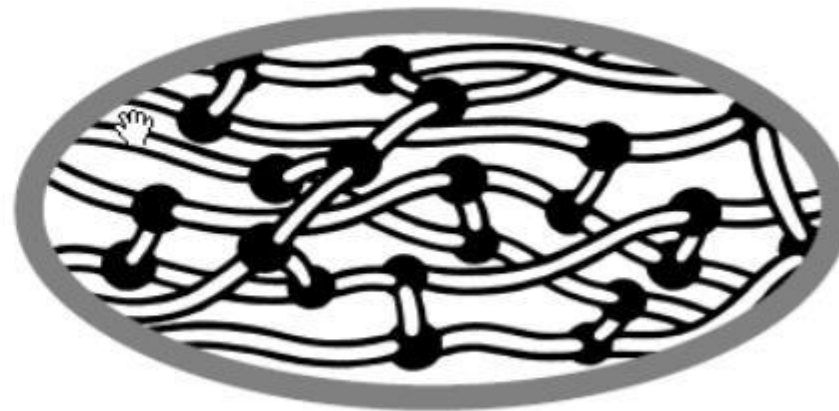
Thermoset

Those which soften on heating and then harden again on cooling



THERMOPLASTIC

Those which never soften once they have been moulded



THERMOSETTING

FORMS OF POLYMERS

Depending on the ultimate form and use, polymers can be categorized as:

(1) Plastic

(2) Elastomer

(3) Fiber

(4) Liquid resin

- A Polymer which can be shaped to a usable Hard and Tough solid form by the Application of Heat and/or Pressure is termed as “**PLASTIC**”.

Examples

- **PE, PP, PVC, POLYSTYRENE, NYLON, etc.**

If a polymer can be drawn into long filamental form such that its length is at least 100 times its diameter, such a polymer product is regarded as a “Fibre”.

Examples:

Nylon, terylene, cellulose, etc.

If the polymers are being used as adhesives, potting compounds, sealants, etc. in liquid form, then such polymers are described as **“LIQUID RESINS”**.

